

Kittichai Panyapa

Unity Gameplay & Multiplayer Programmer

Unity 6 · C# · Photon Fusion 2 · Netcode for GameObjects · Final IK · URP

CONTACT

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SKILLS

Networking
Photon Fusion 2 (Shared Mode), Unity Netcode for GameObjects, Unity Relay, state authority & ownership, [Networked] state, RPC design, networked scene loading

Gameplay
Rigidbody character controllers, ability & interaction systems, puzzle logic, Cinemachine 3, Timeline, Splines, Input System (gamepad)

Animation & Rendering
Final IK, root-motion baking, URP, HLSL & Shader Graph, stencil effects, shader warm-up, MaterialPropertyBlock, profiling

Architecture & Tools
OOP, interface- and event-driven design, ScriptableObject data, Editor tooling, Unity Localization, Addressables

Workflow
Git (GitHub / GitLab), Multiplayer Play Mode testing, daily work with designers & artists

TOGETHER IN THE VOID

PLAYABLE BUILD
hakiren.itch.io/together-in-the-void

GAMEPLAY VIDEO
youtu.be/e-_nx0deeGo

EDUCATION

B.A. Interactive Design & Game
Faculty of Digital Media, Sripatum University
GPA 3.62 / 4.00

PROFILE

Unity programmer focused on co-op games. Sole programmer of **Together in the Void**, a two-player online puzzle-platformer completed end to end: physics character controller, asymmetric co-op abilities, networking, cutscenes, UI, localization and editor tools. I built the multiplayer on **Unity Netcode for GameObjects**, then migrated the whole project to **Photon Fusion 2**, so I am comfortable reasoning about state authority, synchronized state, RPC design and latency across frameworks.

FEATURED PROJECT

Together in the Void · 3D 2-player online co-op, senior thesis — Aug 2025 – Oct 2026

Sole Programmer (gameplay, networking, tools, UI) · 6-person team

Unity 6 (URP) · C# · Photon Fusion 2 · Cinemachine 3 · Timeline · Final IK · Input System · Unity Localization · DOTween

- **Migration to Photon Fusion 2:** ported the entire game from Netcode for GameObjects + Relay to Fusion 2 Shared Mode in six milestones. Each player owns their character; the master client owns world objects through [Networked] state and RPCs. Join-code sessions, synchronized scene loading, and partner-left / connection-lost handling.
- **Netcode feel:** reduced client delay with RPCs that skip tick alignment, change detection on simulation state, and owner-reported zone and hazard events, so each player reacts on their own machine first (32 Hz tick).
- **Character controller (~7k lines):** Rigidbody movement with dash, double jump, coyote time, jump buffering, stair step-up, ledge grab and climb (baked root motion), spline slides, 2.5D side-scrolling, and portal traversal that keeps momentum.
- **Co-op abilities:** Time Freeze (Lyra) and Portal gun (Bobby) on a shared aimed-ability base with per-level unlocks, role checks and a gamepad crosshair that snaps to valid targets; networked portal pairs teleport players, boulders and trains.
- **Animation & IK:** Final IK foot grounding, look-at, aim pose, turn-in-place stepping and ledge hand placement on two Generic (Advanced Skeleton) rigs.
- **Presentation systems:** Timeline and video cutscenes kept in sync on both machines with a hold-to-skip vote; networked dialogue queue with priorities; checkpoint and respawn flow; full Thai/English localization (CSV pipeline + translation editor); Xbox and PlayStation button prompts.
- **Rendering & performance:** custom URP shaders (camera-proximity dither fade, stencil partner silhouette through walls, portal X-ray) and a shader/VFX warm-up during loading that removes first-use hitches.
- **Tooling:** 15+ editor tools, including one-key Solo Play testing, a ledge-reach scanner, a prefab brush, model-to-prefab builders and a humanoid-to-generic clip baker.

250+	~57k	60+	3
C# scripts	lines of code	networked components	levels, full game loop

OTHER PROJECTS

Project SoulSmith · Hades-inspired action combat, personal — 2026 · in development

Hitbox, Hurtbox and Receiver damage pipeline with allocation-free hit payloads and a stagger gauge; inventory and equipment system (ScriptableObject data, serializable runtime items, drag-and-drop UI, stat allocation); weapons and runes swap animation clips through AnimatorOverrideController. Unity 6 · URP · Odin · DOTween.

Computer Assembly Shop Simulator · VR, university — 2025

VR PC-building shop on XR Interaction Toolkit: tag-validated socket assembly for GPU, CPU, cooler and RAM, a trigger-based checkout that totals part prices, and NavMesh customers who request preset builds. GitHub